

1. Introduction

In today's digital world, wireless networks are everywhere — from homes and workplaces to cafes and airports. Understanding how these networks function and how they can be compromised is essential for anyone pursuing a career in cybersecurity. This module, WiFi Hacking 101, provides a foundational look into wireless security, exploring the tools, techniques, and ethical considerations involved in testing the security of WiFi networks. Through hands-on practice with the Aircrack-ng suite and real-world scenarios, learners gain critical skills in packet capture, handshake analysis, and secure network auditing — all within a legal and ethical framework.

2. Module Sections

The screenshot displays the TryHackMe interface for the 'WiFi Hacking 101' room. The browser address bar shows the URL `https://tryhackme.com/room/wifihacking101`. The room progress is indicated as 0% at the top. A green notification box states: 'Woop woop! Your answer is correct'. The main content area contains text explaining WPA and WPA2 security, followed by a section titled 'Answer the questions below'. The first question is 'What type of attack on the encryption can you perform on WPA(2) personal?' with the answer 'bruteforce' entered in the input field. A green 'Correct Answer' button and an orange 'Hint' button are visible. The second question is 'Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)' with the answer 'Nay' entered. A green 'Correct Answer' button is shown. The third question is 'What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?' with the answer '----' entered. A green 'Submit' button is visible. The room progress is updated to 12%.

Previously, the WEP (Wired Equivalent Privacy) standard was used. This was shown to be insecure and can be broken by capturing enough packets to guess the key via statistical methods.

The 4 way handshake allows the client and the AP to both prove that they know the key, without telling each other. WPA and WPA2 use practically the same authentication method, so the attacks on both are the same.

The keys for WPA are derived from both the ESSID and the password for the network. The ESSID acts as a salt, making dictionary attacks more difficult. It means that for a given password, the key will still vary for each access point. This means that unless you precompute the dictionary for just that access point/MAC address, you will need to try passwords until you find the correct one.

Room Banner by [Frank Wang](#) on [Unsplash](#)

Answer the questions below

What type of attack on the encryption can you perform on WPA(2) personal?

bruteforce

✓ Correct Answer

🔍 Hint

Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)

Nay

✓ Correct Answer

What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?

Submit

Room progress (12%)

TryHackMe | Wifi Hacking | What is the three-letter... | https://tryhackme.com/room/wifihacking101 | 150% | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB | OffSec | Academy HTB | TryHackMe | Dashboard | Room progress (18%)

Answer the questions below

What type of attack on the encryption can you perform on WPA(2) personal?

brute force

✓ Correct Answer

Hint

Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)

Nay

✓ Correct Answer

What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?

PSK

✓ Correct Answer

What's the minimum length of a WPA2 Personal password?

—

Submit

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Can this method be used to attack WPA2-EAP handshakes? (Yea/Nay)

Nay

✓ Correct Answer

What is the three-letter abbreviation for the pre-shared key used in Wi-Fi security?

PSK

✓ Correct Answer

What's the minimum length of a WPA2 Personal password?

8

✓ Correct Answer

Task 2 ○ You're being watched - Capturing packets to attack

Task 3 ○ Aircrack-ng - Let's Get Cracking

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Room progress (31%)

You will need a monitor mode NIC in order to capture the 4 way handshake. Many wireless cards support this, but not all. A monitor mode NIC is a network interface card that can be put into monitor mode. In monitor mode, the NIC can capture all traffic on the wireless network, not just traffic destined for the NIC. This is useful for capturing the 4 way handshake, which is a critical part of the WPA2 authentication process. Injection mode helps, as you can use it to deauth a client in order to force a reconnect which forces the handshake normally.

✓ Woop woop! Your answer is correct

Answer the questions below

How do you put the interface "wlan0" into monitor mode with Aircrack tools? (Full command)

airmon-ng start wlan0

✓ Correct Answer

What is the new interface name likely to be after you enable monitor mode?

Submit

What do you do if other processes are currently trying to use that network adapter?

Submit Hint

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Room progress (37%)

How do you put the interface "wlan0" into monitor mode with Aircrack tools? (Full command)

airmon-ng start wlan0

✓ Correct Answer

What is the new interface name likely to be after you enable monitor mode?

wlan0mon

✓ Correct Answer

What do you do if other processes are currently trying to use that network adapter?

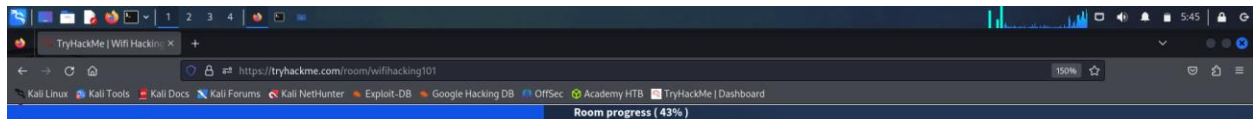
Submit Hint

What tool from the aircrack-ng suite is used to create a capture?

Submit

What flag do you use to set the BSSID to monitor?

Submit



How do you put the interface "wlan0" into monitor mode with Aircrack tools? (Full command)

✓ Woop woop! Your answer is correct

airmon-ng start wlan0

✓ Correct Answer

What is the new interface name likely to be after you enable monitor mode?

wlan0mon

✓ Correct Answer

What do you do if other processes are currently trying to use that network adapter?

airmon-ng check kill

✓ Correct Answer

Hint

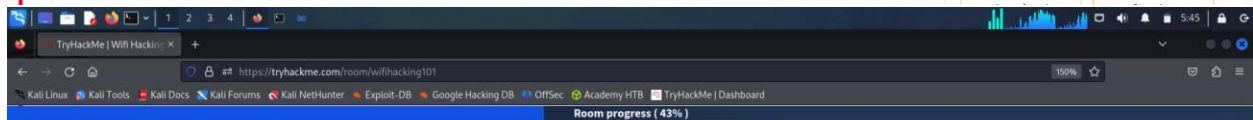
What tool from the aircrack-ng suite is used to create a capture?

airmon-ng

Submit

What flag do you use to set the BSSID to monitor?

airmon-ng



How do you put the interface "wlan0" into monitor mode with Aircrack tools? (Full command)

✓ Woop woop! Your answer is correct

airmon-ng start wlan0

✓ Correct Answer

What is the new interface name likely to be after you enable monitor mode?

wlan0mon

✓ Correct Answer

What do you do if other processes are currently trying to use that network adapter?

airmon-ng check kill

✓ Correct Answer

Hint

What tool from the aircrack-ng suite is used to create a capture?

airodump-ng

✓ Correct Answer

What flag do you use to set the BSSID to monitor?

airmon-ng

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Room progress (56%)

wlan0mon

✓ Woop woop! Your answer is correct

What do you do if other processes are currently trying to use that network adapter?

airmon-ng check kill

✓ Correct Answer

Hint

What tool from the aircrack-ng suite is used to create a capture?

airodump-ng

✓ Correct Answer

What flag do you use to set the BSSID to monitor?

--bssid

✓ Correct Answer

Hint

And to set the channel?

Submit

Hint

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Room progress (62%)

What tool from the aircrack-ng suite is used to create a capture?

airodump-ng

✓ Correct Answer

What flag do you use to set the BSSID to monitor?

--bssid

✓ Correct Answer

Hint

And to set the channel?

--channel

✓ Correct Answer

Hint

And how do you tell it to capture packets to a file?

Submit

Hint

Task 3 ○ Aircrack-ng - Let's Get Cracking

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Room progress (68%)

airodump-ng

✓ Correct Answer

✓ Woop woop! Your answer is correct

What flag do you use to set the BSSID to monitor?

--bssid

✓ Correct Answer

🔍 Hint

And to set the channel?

--channel

✓ Correct Answer

🔍 Hint

And how do you tell it to capture packets to a file?

-W

✓ Correct Answer

🔍 Hint

Task 3 ○ Aircrack-ng - Let's Get Cracking

How likely are you to recommend this room to others?

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Room progress (75%)

ESSID: 'James Honor 8'

✓ Correct Answer

✓ Woop woop! Your answer is correct

Answer the questions below

What flag do we use to specify a BSSID to attack?

-b

✓ Correct Answer

🔍 Hint

What flag do we use to specify a wordlist?

🚩 Submit

🔍 Hint

How do we create a HCCAPX in order to use hashcat to crack the password?

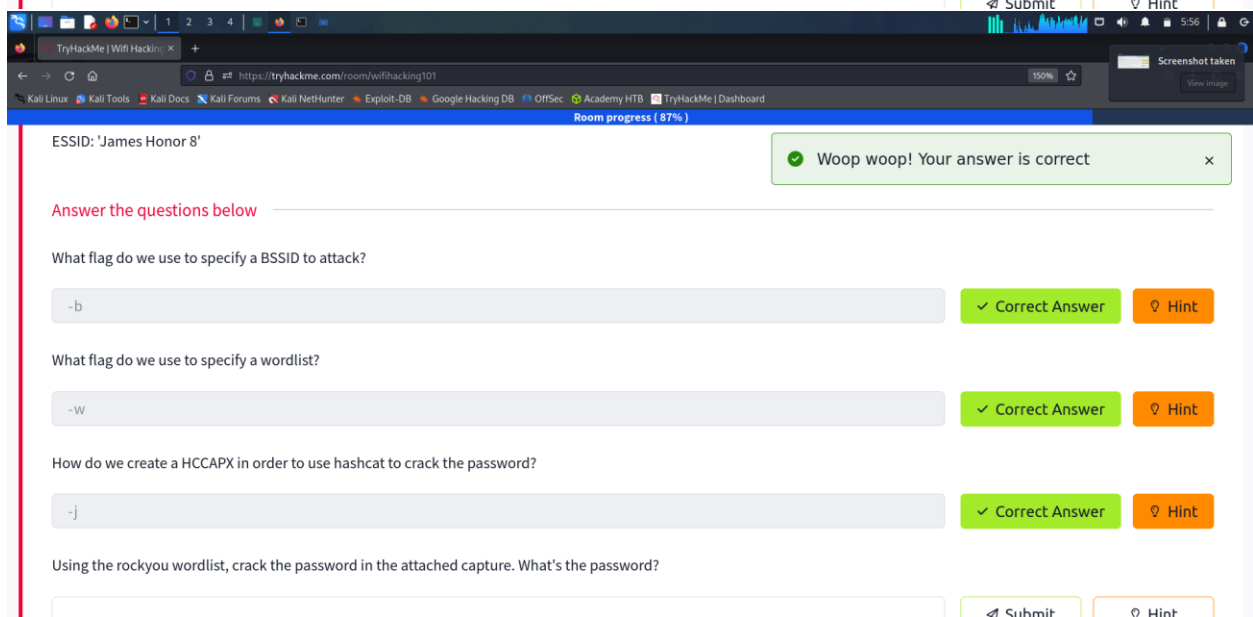
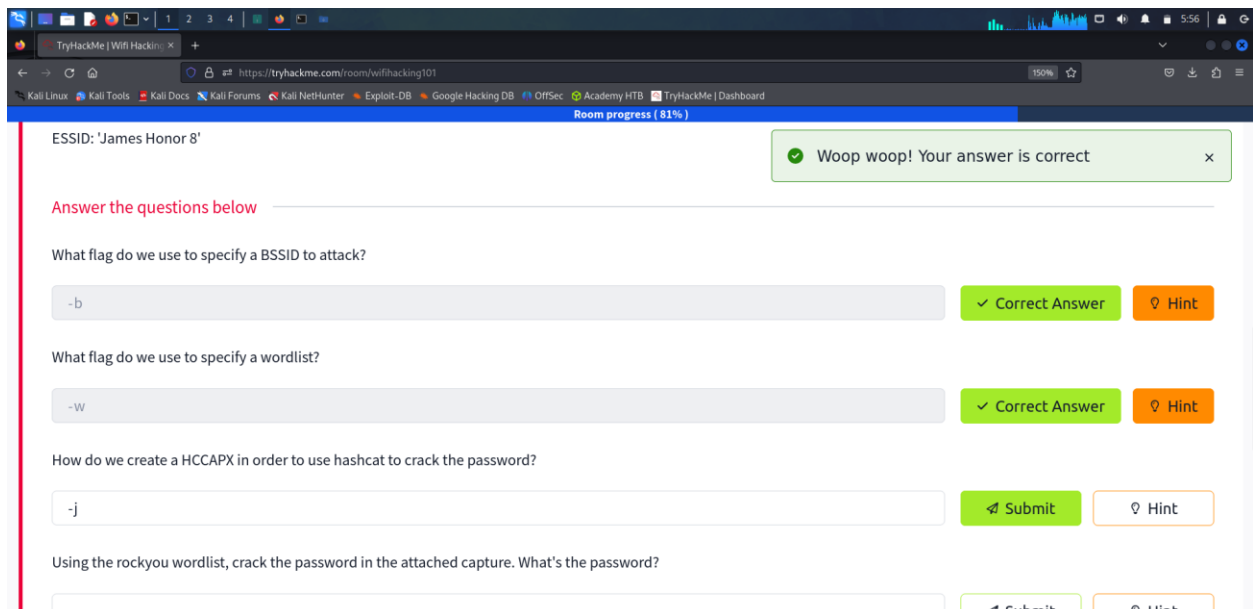
🚩 Submit

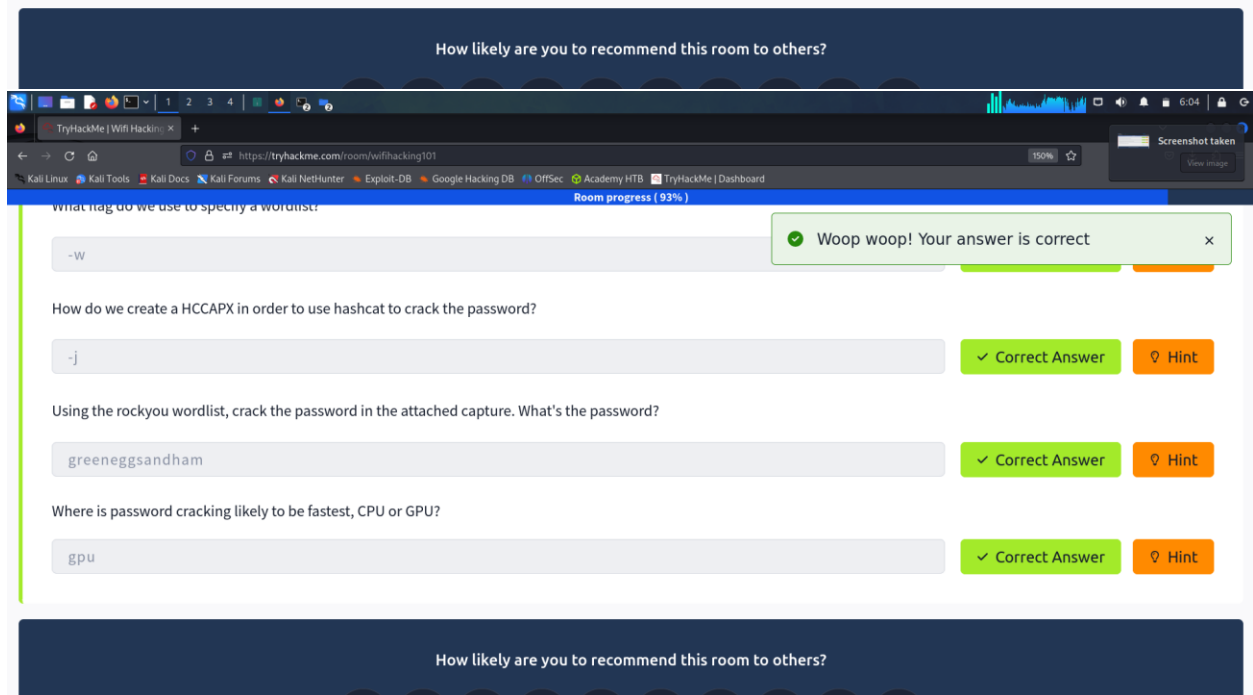
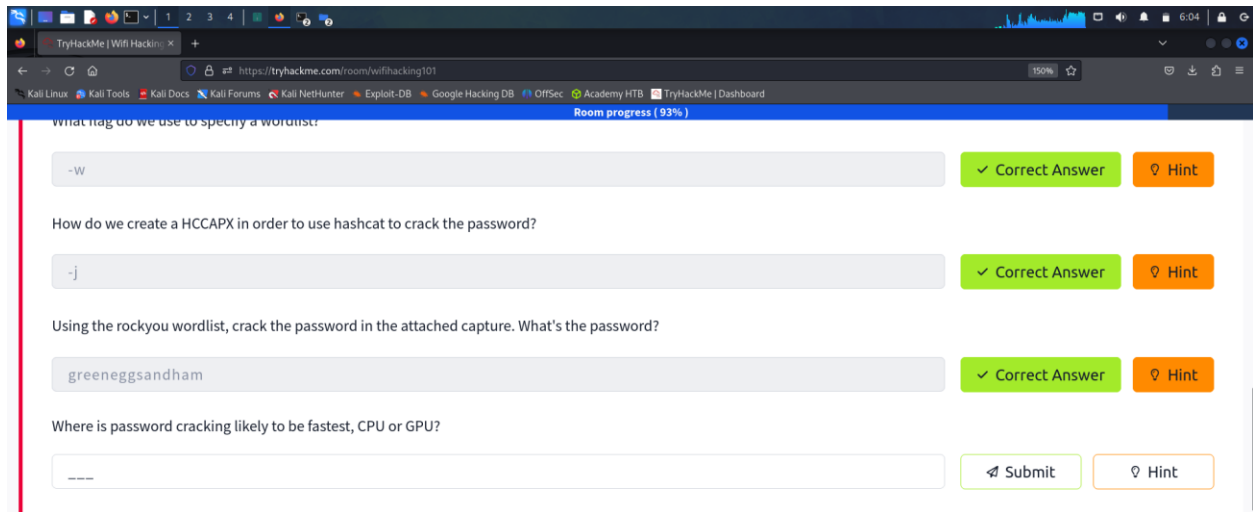
🔍 Hint

Using the rockyou wordlist, crack the password in the attached capture. What's the password?

🚩 Submit

🔍 Hint





3. Shareable Link and Final Screenshot



4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fwifihacking101%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

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https://www.linkedin.com/feed/?shareActive=true&text=Share%20your%20achievement%0D%0Ahttps%3A%2F%2Ftryhackme.com%2Froom%2Fwifihacking101%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

Completing the WiFi Hacking 101 module has been an eye-opening experience that highlights both the power and responsibility that comes with network penetration testing. While it's fascinating to see how easily some networks can be exploited using tools like Airodump-ng and Aircrack-ng, the real takeaway is the importance of securing our own systems — from using strong encryption methods to routinely monitoring for suspicious activity. This module not only provided technical skills but also reinforced the ethical mindset necessary for working in cybersecurity. Moving forward, I feel more confident in my ability to identify and protect against wireless vulnerabilities — and more aware of the vital role ethical hackers play in keeping our digital world safe.